

Two Charts That Hide Their Own Data

One FEC file, drawn as a pie and as a radar, and what each of those shapes does to it

84–355 Democracy’s Data

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Every other chapter in this book picks a form because it suits the data. This one picks two forms because they do not, and then draws them anyway. The file is one that turns up in a pie chart somewhere every election year: the Federal Election Commission’s summary of where each 2024 candidate’s money came from.

The pie chart and the radar chart are the two most widely used graphics in political communication that get harder to read the more carefully you look at them. They fail differently. A pie asks you to compare angles and hides its own total. A radar has an area that depends on a decision nobody writes down.

Both failures can be demonstrated on one file rather than argued about, which is what the rest of this chapter does.

01 One row is a candidate

Source	Dollars	Share
From individuals	\$15,936,794	69.4%
From PACs	\$2,639,351	11.5%
From the party	\$418	0.0%
The candidate’s own money	\$0	0.0%
Transferred in	\$4,384,634	19.1%
Everything else	\$16,000	0.1%

Column	What it holds	Measurement
cand_id	the FEC’s identifier for a candidacy, not for a person	identifier
cand_name	the candidate’s name as filed, in whatever form the committee used	text
ttl_receipts	everything the committee took in during the cycle	continuous
indiv_contrib	money from people, in their own names	continuous
pac_contrib	money from political action committees	continuous
party_contrib	money from the candidate’s own party committee	continuous
self_funding	the candidate’s own money: contributions plus loans, added here	continuous

Column	What it holds	Measurement
trans_from_auth	money moved in from another committee the candidate controls	continuous
other	the receipts left over, computed here as the total minus the five above	continuous

02 Before a pie is possible at all

A pie chart makes three assertions before it draws anything: that the categories do not overlap, that they add to a meaningful whole, and that every one of them is a positive quantity. **This file satisfies none of the three**, and the fixes are the interesting part.

The FEC's categories overlap. `self_funding` is exactly `cand_contrib` plus `cand_loans`, in all 3,856 rows — it is a subtotal sitting alongside its own parts. A pie drawn from the file's fields as they come counts the candidate's own money twice and shrinks every other slice to make room. The identity is checked rather than assumed, and only the subtotal is kept.

They do not add up. The five published sources fall short of `ttl_receipts`, so this chapter works out a sixth category, everything else, as the difference. It is usually small: a median of 0.53% of a candidate's receipts. It is not always small. Thune raised 41.1% of everything outside the five named sources.

Without that residual slice, the one holding what is left over, the pie would stretch the five it has until they filled the circle. The missing 41.1% would then be quietly handed to the other five, in proportion to their size.

Some of the numbers are negative. 14 candidates report negative individual contributions — refunds during the period exceeded receipts, the smallest being \$-118,173. A bar chart draws that below the axis. **A pie chart cannot draw it at all**; there is no such thing as a negative angle in a circle that sums to 360 degrees.

And the whole is not a whole

Candidate ID	Name	Office	State	Receipts
P80000722	BIDEN, JOSEPH R JR	President	00	\$1,175,189,365
P00009423	HARRIS, KAMALA	President	00	\$1,175,189,365
H4AZ07043	GALLEGO, RUBEN	House	AZ	\$64,657,200
S4AZ00139	GALLEGO, RUBEN	Senate	AZ	\$64,657,200
H6MD08549	TRONE, DAVID	House	MD	\$63,833,137
S4MD00319	TRONE, DAVID	Senate	MD	\$63,833,137
H0CA27085	SCHIFF, ADAM	House	CA	\$48,149,197
S4CA00555	SCHIFF, ADAM	Senate	CA	\$48,149,197
H6NV03139	ROSEN, JACKY	House	NV	\$43,915,245
S8NV00156	ROSEN, JACKY	Senate	NV	\$43,915,245

One more thing has to be true before a pie of this file means anything, and it is not: **the**

file contains the same dollars more than once. A person who files for two offices, or whose committee is renamed, gets a candidate record for each, and the FEC reports the committee's totals under every one of them.

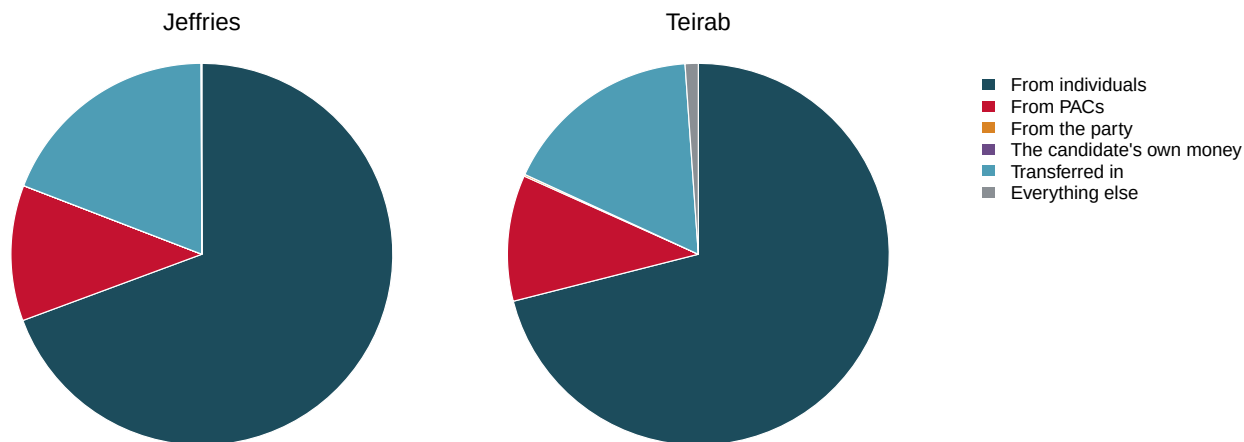
313 records in 142 groups report receipts identical to the cent. The largest is the largest campaign in the file: BIDEN, JOSEPH R JR and HARRIS, KAMALA each carry \$1,175,189,365, because Biden's committee was renamed for Harris and both candidate IDs inherited its totals. The widest is Ekpete Kama, who appears under 8 separate candidate IDs with \$54,191 each.

Added up as filed, those 313 records hold \$3.14 billion. Counted once each, they hold \$1.57 billion. **A chart of "all money raised in 2024" built from this file without checking contains \$1.57 billion that does not exist.** A pie is the chart least able to show it, because a pie has no axis on which a total can be checked. This chapter drops 171 duplicate records and works with 3,685.

Before you look at the figures, commit to a guess. Two campaigns below are drawn as pies. One of them raised about nine million dollars and the other about sixty-seven million. **Looking only at the two circles, how would you tell which is which?** Write down what you would look for.

03 The pie

The same numbers as pies



The same numbers as bars, in dollars. Solid: Jeffries, \$23.0m. Pale: Teirab, \$3.3m.

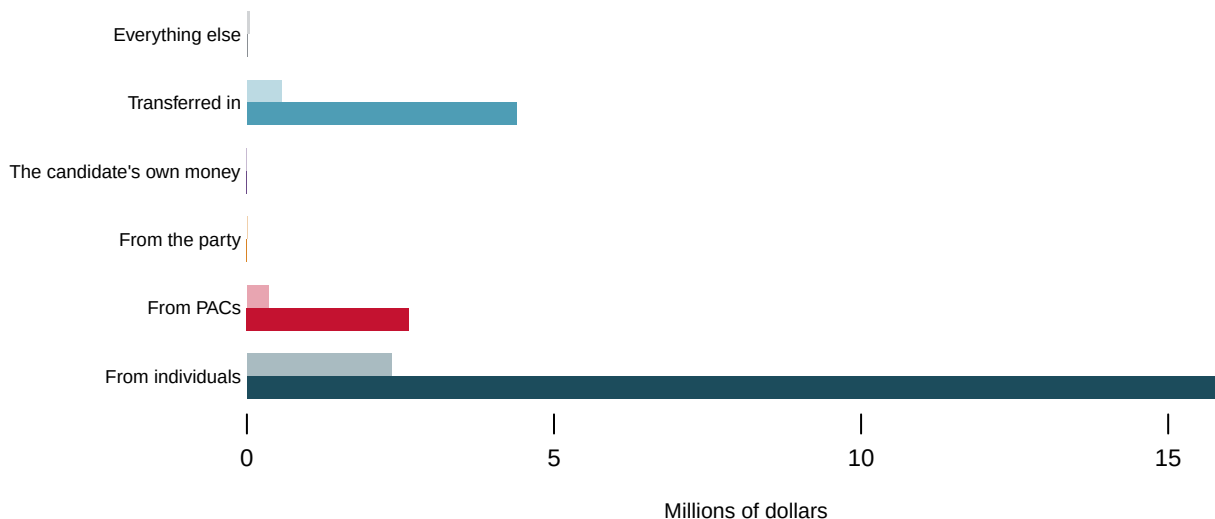


Figure 1. Two campaigns as pies, and the same numbers as bars. The circles are Jeffries and Teirab. In print both panels are shown; on screen the button switches between them.

The two pies are the same picture. No slice differs from its counterpart by more than **2.06 percentage points** — a difference no eye can resolve at any size. And **one of these campaigns raised 7.0 times as much as the other**: \$23.0m against \$3.3m.

That is not a distortion. It is the pie chart working exactly as designed. A pie is drawn from shares and its radius carries no information. **Two pies are never comparable unless somebody writes the totals underneath**, and at that point the reader is doing the comparison from the text, with the circles as decoration.

Now press the button, or look at the lower panel. The bars are on a shared dollar axis and the question answers itself.

The second failure is inside a single pie. Try to rank Teirab's three smallest sources from

the circle alone. You cannot, and it is not your fault. Cleveland and McGill (1984) measured how accurately people read graphical encodings and found a clear order: position along a common scale first, then length, then angle and area near the bottom. **A pie uses two of the worst three.** The bar chart uses the best one.

04 The radar

A radar chart, also called a spider or star chart, puts each variable on its own spoke, plots the value along it, and joins the points into a polygon. It looks like it solves the problem the pie has, because it can hold several entities at once and it shows magnitudes rather than shares.

It introduces a worse one.

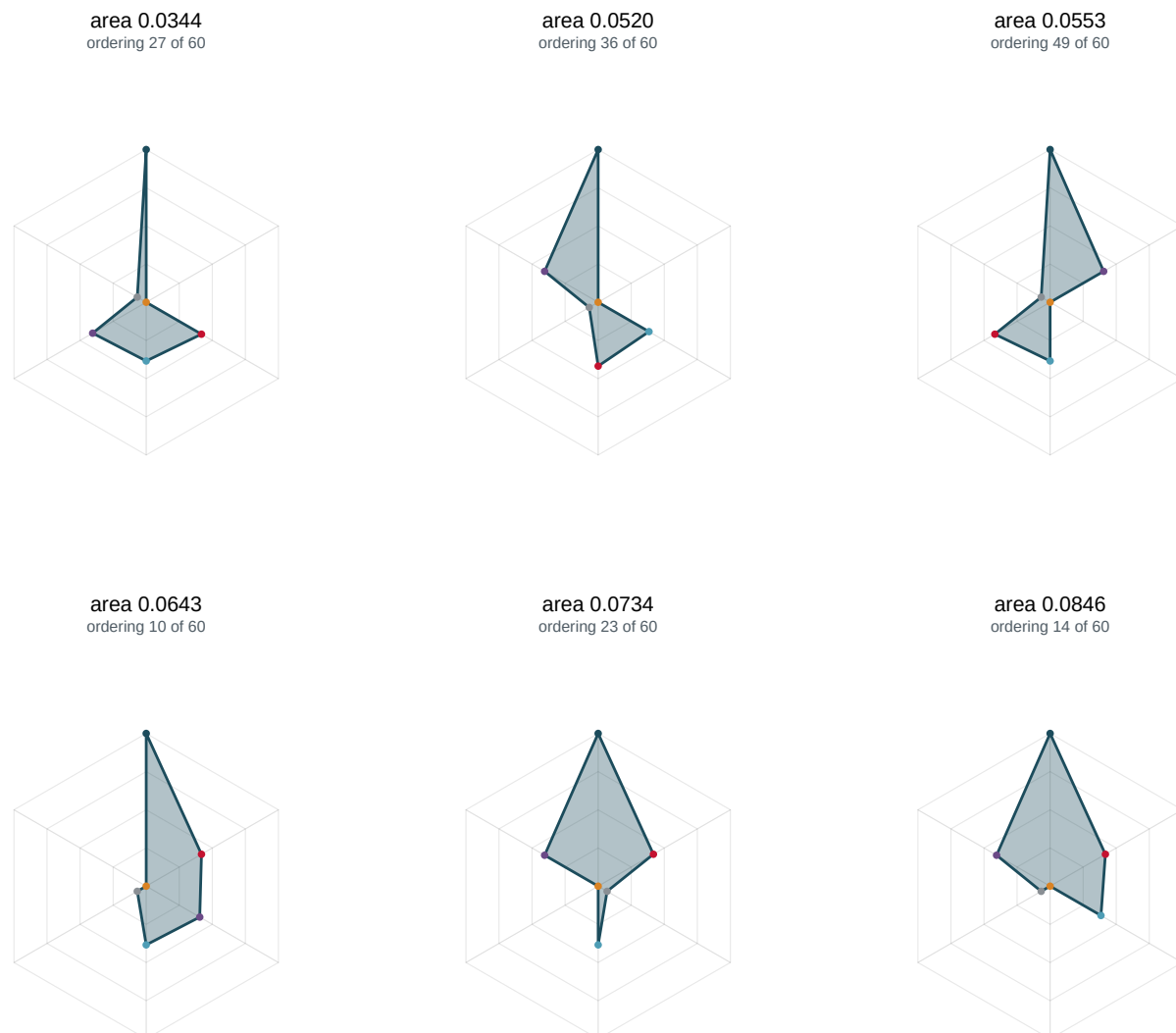


Figure 2. One candidate, six axes, and every way of ordering them. Pick a campaign with the buttons. **Reorder the axes** steps through all 60 distinct arrangements of the six spokes — not a single number changes, and the polygon does. The strip on the right marks the area of every arrangement, with the current one in red. In print, six of the 60 arrangements for Steel are shown side by side.

Press it a few times. **The area of the shape is a property of the order the axes were put in**, and that order is a decision the chart never records. The arithmetic is not subtle: for 6 axes at equal angles with radii r_i , the area is

$$A = \frac{1}{2} \sin\left(\frac{2\pi}{n}\right) \sum_i r_i r_{i+1}$$

— every term is a product of **adjacent** radii. Put the two largest values next to each other and the area is large; separate them and it collapses. The axes here are funding sources, and there is no sense in which PAC money comes “after” party money, so every one of the 60 arrangements is equally defensible.

Campaign	Profile	Smallest area	Largest area	Ratio
Harris	almost all from individuals and transfers	0.0000	0.1080	3947×
Trone	almost all the candidate’s own money	0.0000	0.0064	12701×
Scalise	almost all transferred in	0.0001	0.0653	1115×
Smith	the most PAC-funded campaign in the file	0.0002	0.1077	496×
Steel	the most evenly spread profile in the file	0.0344	0.0846	2.46×

The sparse profiles are the extreme cases. Trone raised almost everything from one source, so five of the six radii are near zero. The polygon is either a thin triangle or a sliver, depending on where the one big spoke lands. Ignore them. **Look at Steel**, the most evenly spread profile in the entire file — money on five of the six axes, the case a radar chart is supposed to be good at. The same six numbers make polygons from 0.0344 to 0.0846: a factor of **2.46**, on the best-behaved candidate available.

The second button is the other half of the problem. By default each axis is scaled to the largest of the six values, which is what makes the polygon fill the circle. Press it and every axis is put on one dollar scale, and the shape collapses toward the one or two axes carrying real money. **Neither is wrong, and they are different pictures of identical numbers.** A radar chart cannot tell you which one it is showing you, because it has no axis labels with units on them — a spoke is a direction, not a scale.

A second prediction. Suppose a seventh axis were added — say, receipts from party committees split into national and state. **What happens to the area of every polygon in Figure 2, before any candidate raises or spends a dollar differently?** Say whether it goes up or down, and why the answer has nothing to do with campaign finance.

05 When each of these is fine

Neither form is banned, and saying so matters more than the criticism.

A pie is defensible when there are two or three categories, they genuinely partition one whole, the whole is stated in words nearby, and nobody is being asked to compare it to another pie. “Fifty-two per cent to forty-eight” is a fact a circle can carry. Six slices, two circles and a comparison is three things past that line.

A radar is defensible when the axes have a natural cyclic order that the reader already knows — months of the year, compass directions, hours of the day. Then the adjacency the area depends on is not arbitrary, because the data supplied it. Funding sources have no such order. Neither do the “candidate quality” and “state profile” radars that appear every cycle.

What to draw instead, in both cases, is in the lower panel of Figure 1: one bar per category, on a common scale, sorted by size. It is the least interesting graphic in this chapter and it answers every question the other two cannot.

06 What this chapter cannot tell you

This is a chapter about drawing, not about money. Which candidates raise from whom, and what that buys, is the campaign-finance chapter's subject, and the network of outside spending is finance-network's. Nothing here should be read as a finding about American campaign finance.

The sixth category is a residual, and a residual is not a source. “Everything else” is `ttl_receipts` minus five published fields. It contains offsets to operating expenditures, loans from other people, and receipts the summary file does not itemise, which are different things with different meanings. It is drawn as one slice because a pie needs the parts to close, and that is exactly the kind of convenience this chapter is about.

The duplicate rule is strict and therefore undercounts. 142 groups match to the cent across every receipt field. Two records for the same person that differ by a dollar are not caught. That happens when they were pulled at different times, or one committee filed an amendment. The \$1.57 billion of double-counted money is a floor.

Cleveland and McGill measured accuracy, not usefulness. Their ranking is about how precisely people read a value off an encoding. A pie is read accurately for one thing, which is whether a share is above or below a half, and that is a real question in politics. The criticism here is of pies asked to do more than that, which is most of them.

Sixty orderings is exact, and six axes is a choice. With 6 categories there are 60 distinct cyclic arrangements, enumerated rather than sampled. Split or merge a category and the count changes, and so does every area in Figure 2. That is the same point one level up: the arbitrariness is not only in the order of the axes, but in there being six of them.

07 Sources

- Federal Election Commission. *Candidate summary file, 2023–2024 cycle*, as captured and committed by this corpus's campaign-finance chapter. The figures here read that chapter's published output, check the overlap between `self_funding` and its two components, compute the residual category, and enumerate the duplicate candidate records. <https://www.fec.gov/data/browse-data/?tab=bulk-data>
- Cleveland, William S. and Robert McGill (1984). “Graphical Perception: Theory, Experimentation, and Application to the Development of Graphical Methods.” *Journal of the American Statistical Association* 79(387): 531–554. The experiments behind the ranking of encodings used above.
- Spence, Ian and Stephan Lewandowsky (1991). “Displaying Proportions and Percentages.” *Applied Cognitive Psychology* 5(1): 61–77. The careful defence of the pie chart for

the narrow task it is good at.

- Playfair, William (1801). *The Statistical Breviary*. London. The first published pie chart, of the territories of the Turkish Empire — two categories, one whole, no comparison.
- Saary, M. Joan (2008). "Radar plots: a useful way for presenting multivariate health care data." *Journal of Clinical Epidemiology* 61(4): 311–317. A sympathetic account of the radar chart, including the axis-order problem.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`areas.csv`, `candidates.csv`, `duplicates.csv`, `featured.csv`, `lookalikes.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `candidates.csv` has each source of money as both a dollar amount and a percentage. Pick a candidate and work out which slices would be too thin to see on a pie. What disappears?
- `areas.csv` computes the area of the same radar chart under different orderings of the axes. Compare them for one candidate. Should the order of the axes change the size of the shape?
- `lookalikes.csv` holds candidates whose charts look alike. Compare their actual dollar figures. Is the resemblance real?
- `duplicates.csv` groups records that refer to the same candidate. Count them. What happens to any total computed before the duplicates are found?